The problem here is to find the HCF of 18 and 27.

Prime Factors of 18 = 2 x **3** x **3**

Prime Factors of 27 = 3 x **3** x **3**

Common prime factors of 18 and 27 = 3 x 3 = 9

So, the HCF of 18 and 27 is **9.**

Now, let us learn some more methods of finding the HCF. The largest team member each group is **9**, Std VII has 2 teams and Std VIII has 3 teams.

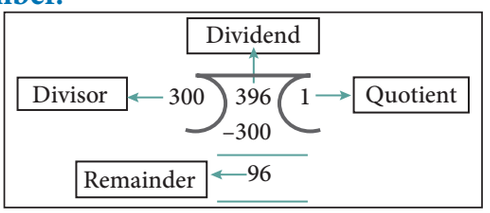
3. Repeated Division Method:

The above methods are easy to finding HCF, but for larger numbers these methods are tedious to find factors of the given numbers. In that case, alternatively we have some more methods to find HCF. Let us learn more about the other methods of finding the HCF.

For the above Situation, what if the Class VII had 396 students and Class VIII had 300 students? Then, what would be the biggest possible team size? Well, the above said two methods may not help us quickly. So, we can use continuous division method for finding the highest common factor.

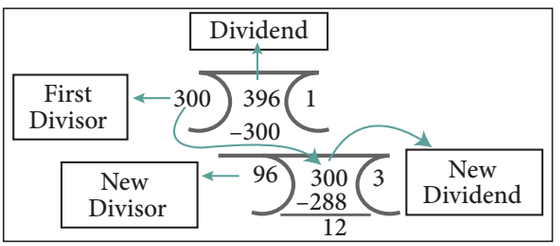
STEP 1: Divide the larger number by the smaller number.

Here, 360 is the larger number. So, we divide 360 (Dividend) by 300 (Divisor). We get the Remainder as 96.



STEP 2: The remainder from Step 1 becomes the new divisor, and divisor of Step 1 becomes the new dividend.

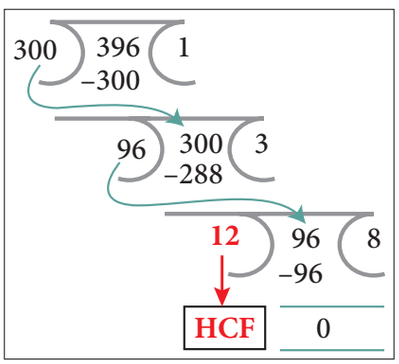
From the step 1, we got 96 as remainder. So, in the second step 96 becomes the new divisor and 300 becomes the new dividend.



STEP 3: Repeat this division process till remainder becomes zero. The divisor of the last division (when remainder is zero) is the HCF.

From step 2, we got 12 as the new remainder which will become the new divisor. In the third step 12 becomes the new divisor and 96 becomes the new dividend. **Now, the remainder is zero when 12 is the last divisor of the division. Therefore,12 is the required HCF.**

Hence, the **HCF of 396 and 300 is 12.** So each team would be **12 students.**



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